SNOWPLOW MODELING OF A LONG-CONDUCTION-TIME PLASMA OPENING SWITCH

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Abstract

The modified snowplow model of plasma opening switch is proposed. The conduction phase is divided into two parts. The first part lasts until the current-carrying channel reaches the load end of the pre-filled plasma. The second part, the transition phase to the opening, is characterized by the motion of current carrying channel downstream the initial plasma location. The equation of motion of the current channel is solved for both phases and the snowplow mechanism for high-voltage pulse generation is developed. The theoretical dependencies are in good agreement with the experiment, which includes interferometry and optical fiber measurements of current front translation. The qualitative model of the switch plasma behavior is proposed and used in explanations of various plasma opening switch features.

I. INTRODUCTION

The major difference between nanosecond plasma opening switch (POS) and its microsecond counterpart is in different plasma density, which is necessary to achieve high conduction current. Higher plasma density in a microsecond, long-conduction-time opening switch results in different processes taking place during both conductive and opening phases. If ion erosion process is considered as predominant in the operation of a nanosecond POS [1], the snowplow model is found to play main role in a microsecond POS [2].

The snowplow model of the plasma displacement in a microsecond POS assumes weak magnetic field penetration into the bulk plasma. The model describes well existing POS scalings [3] in a wide range of switch parameters. The model does not require the magnetic field penetration into the bulk plasma and it is free of assumptions like anomalous plasma resistivity and shock-wave-like field penetration.

In the pioneering work [2] the plasma evolution in the frame of the snowplow model is divided into two distinct phases. For the conduction phase a snowplow criterion was obtained as a function of current, time, plasma density and length and electrode radii. The plasma develops a thin current carrying channel, which propagates along the switch in the direction of the load. The important point obtained in this study is that the snowplow model is the only approach to obtaining hydrodynamic scaling for the conduction current.

However, in this model, the mechanism of the transition from the conduction phase to the opening phase is unclear. The major switch phases of operation are treated separately. The authors of [2] mentioned that the model might result in correct scaling even if snowplow represents incorrectly the detailed hydrodynamic processes in the switch plasma. At the time when the paper [2] was published, the plasma density measurements by laser interferometry were not available. Recently, this diagnostics have been used in numerous studies and provide new important information on the POS physics.

This paper is dedicated to further development of snowplow model of the operation of a microsecond plasma opening switch. Newly obtained data as well as literature data allowed suggesting the modifications to the model for a better agreement with experimental facts.

II. CONDUCTION PHASE

Taking the assumption standard for the snowplow POS model [2] and assuming that the current through plasma I rises linearly, equation of motion for the current-carrying channel is

$$I = I_0 t;$$
$$M_i n \frac{d(lv)}{dt} = \frac{\mu I_0^2}{8\pi^2 r^2} t^2. \quad (1)$$

Here l is the variable length of switch changing in the process of the snowplow motion, M_i is the average ion mass, v is the snowplow velocity, n is the plasma density and r is the plasma radius. The right-hand side of the equation of motion represents Ampere's force.

Designating

$$c_1 = M_i n; \; c_2 = \frac{\mu I_0^2}{8\pi^2 r^2}.$$

the equation

$$c_1 \frac{d(lv)}{dt} = c_2 t^2 \quad (2)$$

can be solved dividing the variables. Therefore, the current-conducting layer moves with constant acceleration:

$$a = \sqrt{\frac{\mu I_0^2}{12\pi^2 r^2 M_i n}} \quad (3)$$

and its velocity linearly rises with time.

III. TRANSITION PHASE – PLASMA THINNING

After the current channel leaves the pre-formed plasma, its mass does not rise anymore and, therefore, the equation of motion has a different form. If the current is

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14. ABSTRACT
The modified snowplow model of plasma opening switch is proposed. The conduction phase is divided into two parts. The first part lasts until the current-carrying channel reaches the load end of the pre-filled plasma. The second part, the transition phase to the opening, is characterized by the motion of current carrying channel downstream the initial plasma location. The equation of motion of the current channel is solved for both phases and the snowplow mechanism for high-voltage pulse generation is developed. The theoretical dependencies are in good agreement with the experiment, which includes interferometry and optical fiber measurements of current front translation. The qualitative model of the switch plasma behavior is proposed and used in explanations of various plasma opening switch features.

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close to its maximum, it can be assumed to be constant during the end of the conduction phase and during the opening phase. The equation of motion becomes therefore:

$$M_i nl\frac{dv}{dt} = \frac{\mu I^2}{8\pi^2 r^2}. \quad (4)$$

The channel is moving with a constant acceleration.

$$v = v_0 + at;$$

$$a = \frac{\mu I^2}{8\pi^2 r^2 M_i nl}. \quad (5)$$

Where v_0 is the velocity, which the channel obtained during the motion in the pre-formed plasma, during the first half of the conduction phase.

Therefore, there are two different phases within the phase of conductivity in a microsecond POS. The first phase lasts until the current-carrying channel reaches the load end of the pre-filled plasma. The second phase, the transition phase to the opening, is characterized by the motion of current-carrying channel downstream the initial plasma location.

According to the model developed for a microsecond POS, a very important phase for a POS opening is the plasma thinning. The decrease in the plasma density, resulting finally in the opening, in the second half of the conduction phase was attributed to the beginning of the radial motion of the plasma outward of the curved current channel. Corresponding decrease in the line-integrated plasma density was obtained in numerous interferometer measurements.

The degree of the plasma thinning is strongly dependent on the curvature of the current-conducting channel. Direct measurements of the magnetic field penetration into the microsecond POS [4] showed mostly radial current channel during a substantial part of the conduction phase making plasma thinning impossible. Therefore, the plasma thinning and transition between conduction and opening phases in a long-duration POS occurs mostly after the current-conducting layer leaves the pre-filled plasma. Starting from that time, the channel becomes not radial [4] and plasma may move in radial direction due to axial component of the plasma current resulting in "plasma thinning" – decrease in the plasma density during the transition phase from the conduction to the opening.

The ratio l/d where l is the length of the switch plasma and d is the radial distance between electrodes determines features of plasma thinning process. If the ratio l/d is small the current conduction channel is mostly radial and the plasma thinning occurs after the current-carrying channel leaves initial plasma location. In the opposite case (large l/d) the shape of the current carrying channel becomes important and the snowplow propagation inside the pre-formed plasma may result in the plasma density decrease. Noteworthy, that low l/d ratio is realized in most cable gun experiments, high l/d ratio is, probably, a good approximation for the flashboard experiments.

IV. OPENING PHASE

Traditionally, the voltage generated during the POS opening is considered as an inductive voltage drop due to fast decrease/increase of the generator/load current I_G / I_L. However, if the current conducting channel is rapidly accelerating downstream, the voltage is generated due to the change of inductance, like in a Z pinch. Assuming a constant acceleration of the current-carrying channel for the case of nearly constant generator current, the total generator inductance can be written as:

$$L = L_0 + L'(v_0 + \frac{at^2}{2}) \quad (6)$$

Taking into account that v_0 is comparatively small and the magnetic flux is conserved (generator current close to its maximum and nearly constant in time) the time behavior of the generator current is:

$$I(t) = \frac{\Phi_0}{L(t)} = \frac{L_0 I_0}{L_0 + L'\frac{at^2}{2}}; \quad (7)$$

The POS voltage (more exactly, the voltage between POS electrodes at the initial plasma location) can be calculated then as:

$$V_{POS} = L_0\frac{dI(t)}{dt} = L_0\frac{d\left(\dfrac{L_0 I_0}{L_0 + L'\frac{at^2}{2}}\right)}{dt} = -\frac{L_0^2 I_0 L' at}{\left(L_0 + L'\frac{at^2}{2}\right)^2} \quad (8)$$

This dependence has a maximum, which occurs at the time calculated from the zero derivative. The solution results in four order algebraic expression with very bulky, and useless roots. However, taking into account that

$$L_0 >> L'\frac{at^2}{2} \quad (9)$$

at low t (t, here, is the duration of the conduction-opening transition phase, which is in proportion to the duration of the POS conduction phase according to the experimental data given in Section VI) we obtain

$$V_{POS} \approx -L_0 I_0 L' at \quad (10)$$

Substituting from Eq. (5)

$$a = \frac{\mu I^2}{8\pi^2 r^2 M_i nl}; \quad \therefore n \propto IT \quad (11)$$

We find the final scaling for the POS voltage.

$$V_{POS} \propto \frac{I^2}{T r^2} \quad (12)$$

Here, I is the POS current, and T is the conduction time. This scaling is obtained from snowplow model without assumption [2] that the snowplow changes its direction from axial to radial with the transition from the conduction to the opening phase.

V. SIMPLIFIED PICTURE OF POS OPERATION

The simplified picture of the POS operation according to our experimental and theoretical analyses is shown in Fig. 1. This sequence represents the case where the plasma thinning occurs after the current carrying channel leaves the pre-formed plasma. The pictures correspond to the respective arrows on the Fig. 2, which contains typical waveforms of load and generator currents and plasma density behavior in time.

Fig. 1(a) shows the pre-filled plasma.

Fig. 1(b) corresponds to a standard snowplow conduction phase with preliminary radial current-conducting channel.

Fig. 1(c) shows the phase, which corresponds to the plasma thinning phase in the conventional [2] snowplow model. However, there can be no thinning in this phase as it is mentioned above. The degree of thinning must be strongly dependent on the POS length/gap ratio.

Fig. 1(d) shows the current layer, pushing the plasma, leaves the region of initial plasma fill.

Fig. 1(e) shows the phase of the plasma thinning where the current has both radial and axial components. The current layer accelerates toward the load loosing some plasma during its motion.

Fig. 1(f) corresponds to a complete separation of the dense pre-filled plasma and comparatively low-density plasma at the current conducting channel. It is seen how large gap in dense plasma can develop without a formal opening, e.g. there is no current through the load.

Fig. 1(g) corresponds to the opening when the load current monitor shows the increase in the load current. The interferometer can still detect some plasma density resulting in the incorrect conclusion that no vacuum gap is formed in the POS. Another indication of the POS opening is the decrease of the generator current. This decrease can be due to fast increase of the generator circuit inductance during the current channel motion while the magnetic flux Φ=LI is nearly conserved.

Fig. 1(h) correspond to the plasma recombination at the window and load endplate. The interferometer may show fast decrease in the plasma density at this moment. The measured gap can be very large because its expansion started long time ago at the time corresponding to Fig. 1(g).

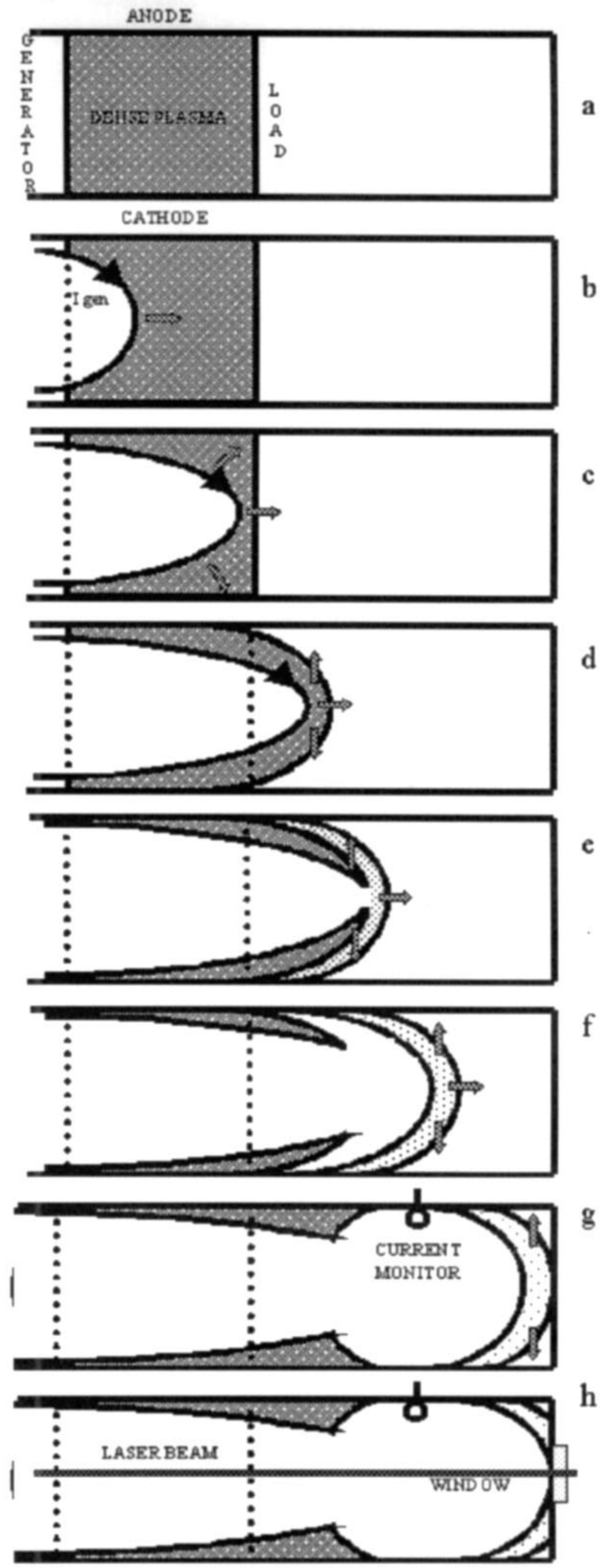

Fig. 1. Model of POS operation. Description in text.

VI. EXPERIMENT

The temporal behavior of the plasma density in a shot is shown in Fig 2. The arrows indicate the time moments during the switch operation, which correspond to the sketches in Fig. 1, which will be discussed in the following sections. The generator and load waveforms are standard for POS experiments.

The plasma density waveform can be divided into several phases. Initially, the POS plasma density follows the dashed line, which indicates the plasma density trace without current in the POS. Then, between points b and c the line-integrated plasma density rises over plasma gun density value. Later, the plasma density decreases monotonously reaching zero at the opening.

The ionization degree of the plasma and its average charge state are lower in the plasmas produced by the cable plasma guns and the density increases due to the ionization process. The ionization of the plasma atoms/ions occurs, probably, at the current front. Assuming constant degree of ionization along POS plasma and taking into account that increase of the line integrated (along z-axis) electron density is proportional to the path of the current layer in plasma $\Delta n \propto l$. Knowing that the current-carrying channel moves with constant acceleration from Eq. (3) within the pre-formed plasma, we obtain $\Delta n \propto t^2$. This dependence corresponds well to the experimental data.

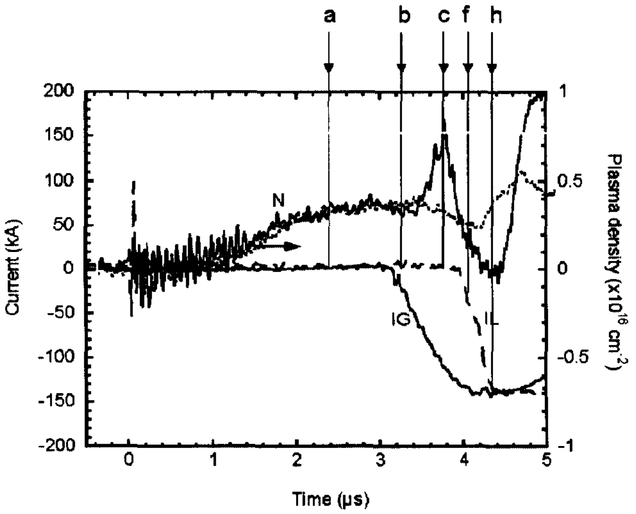

Fig. 2. Experimental waveforms.

After the plasma density reaches its maximum, it decreases monotonously down to zero. Minimum density is observed in the end of the switching process. This is plasma thinning, which corresponds to sketches in Fig. 1(d-g).

Minimum plasma density often corresponds to the opening of the switch – moment when the load current monitor shows fast increase of the current. However, the time corresponding to the lowest plasma density can be delayed. The way in which this is possible shown in Fig. 1(g). If either the downstream part of the POS hardware is considerably long or only a small amount of the plasma is accelerated in the conduction-opening transition process, the opening and zero-density signal from an interferometer may coincide in time.

An additional experiment was carried out with the optical fibers measuring current front translation in time. Assuming the plasma gun position to be zero on the axis of the coaxial POS electrode system, in the first experiment the fibers were placed in the axial positions – 15, 0, 15 and 30 mm (i.e. within the pre-formed plasma). In the second experimental run all fibers were moved in the downstream direction and their locations became 20, 50, 80, 110 and 140 mm, outside of the pre-formed plasma except the first, 20 mm location.

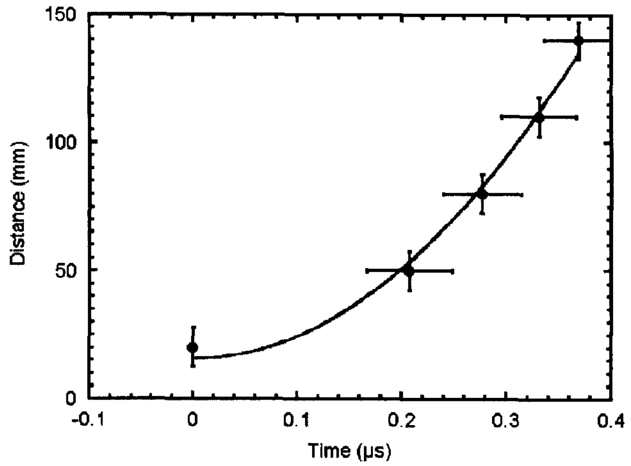

Fig. 3. Propagation of current-conducting channel in downstream POS region.

The time behavior of the current conducting layer during the transition conduction-opening phase is shown in Fig. 3. This dependence of the current front location versus time is well represented by a quadratic power function in both conduction and transition phases, which is in good agreement with the models developed in previous sections.

Most of experimental dependencies [4-6] can be explained in the frame of our modified snowplow POS model. It explains increase of the density in the first half of the conduction phase and its decrease in the second half. From our model, it becomes clear why plasma density rapidly decreases at/after opening and why a large gap can be formed. Model explains also experimentally found dependencies of POS parameters on the value of downstream inductance and on plasma ion species.

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